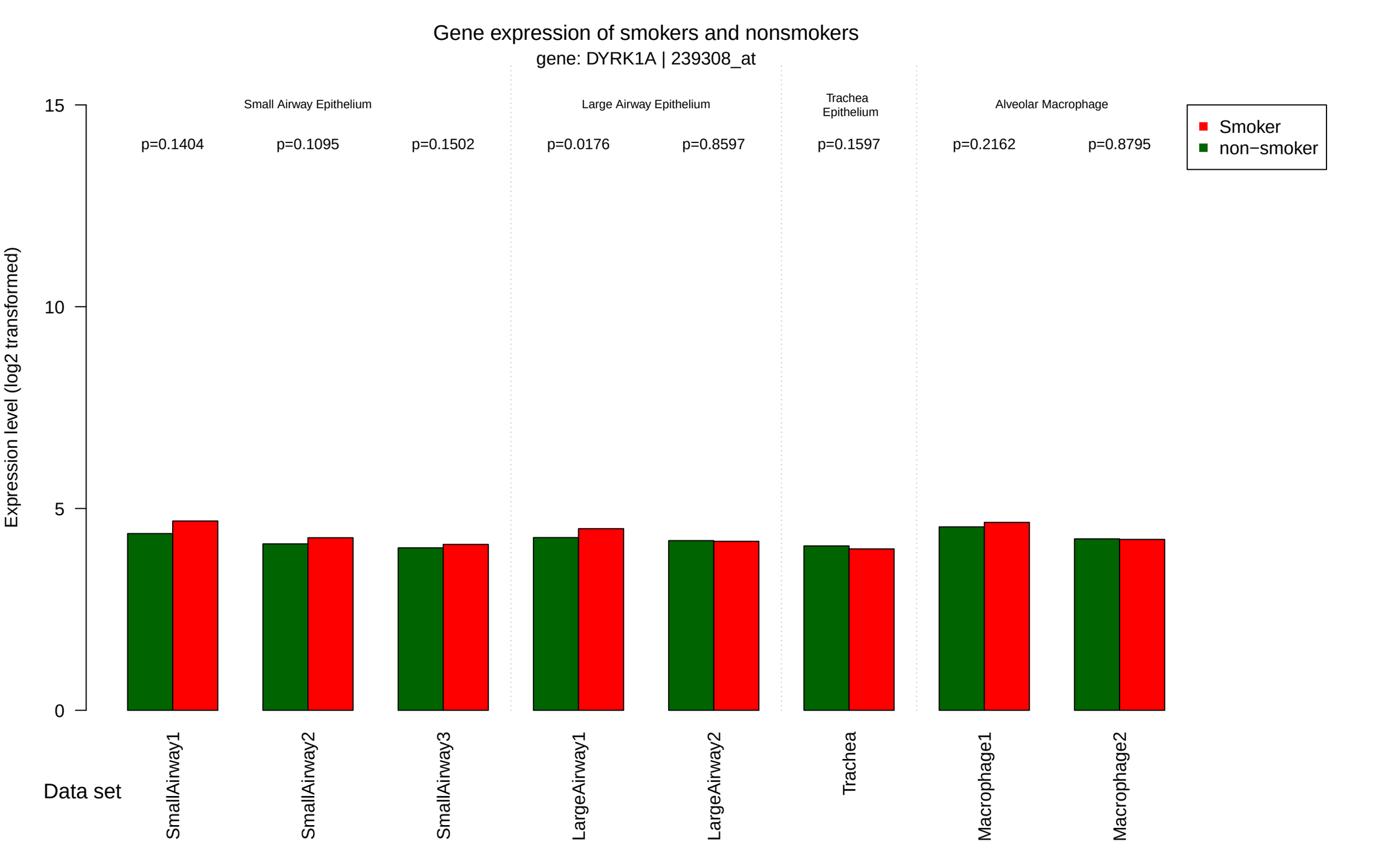
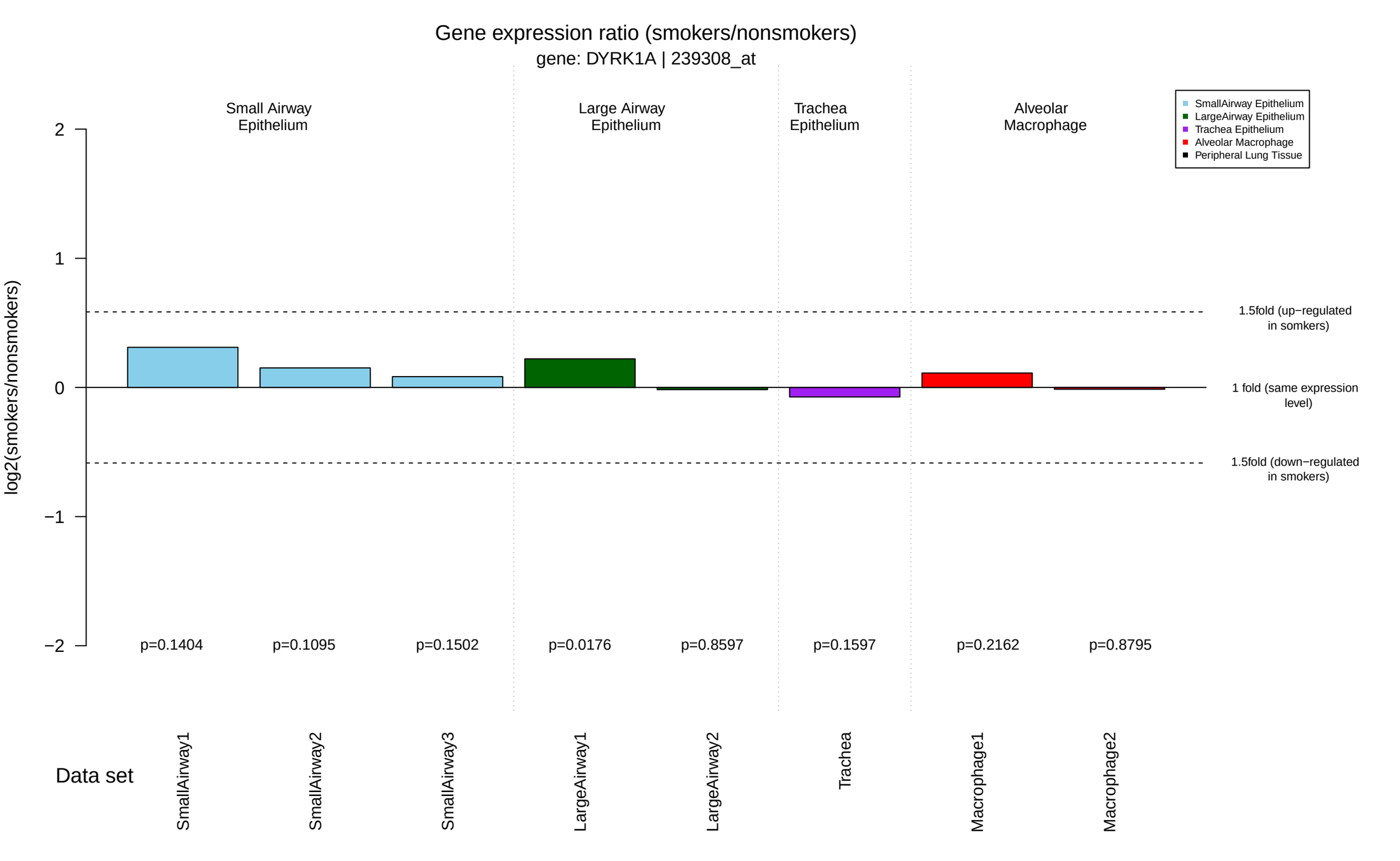
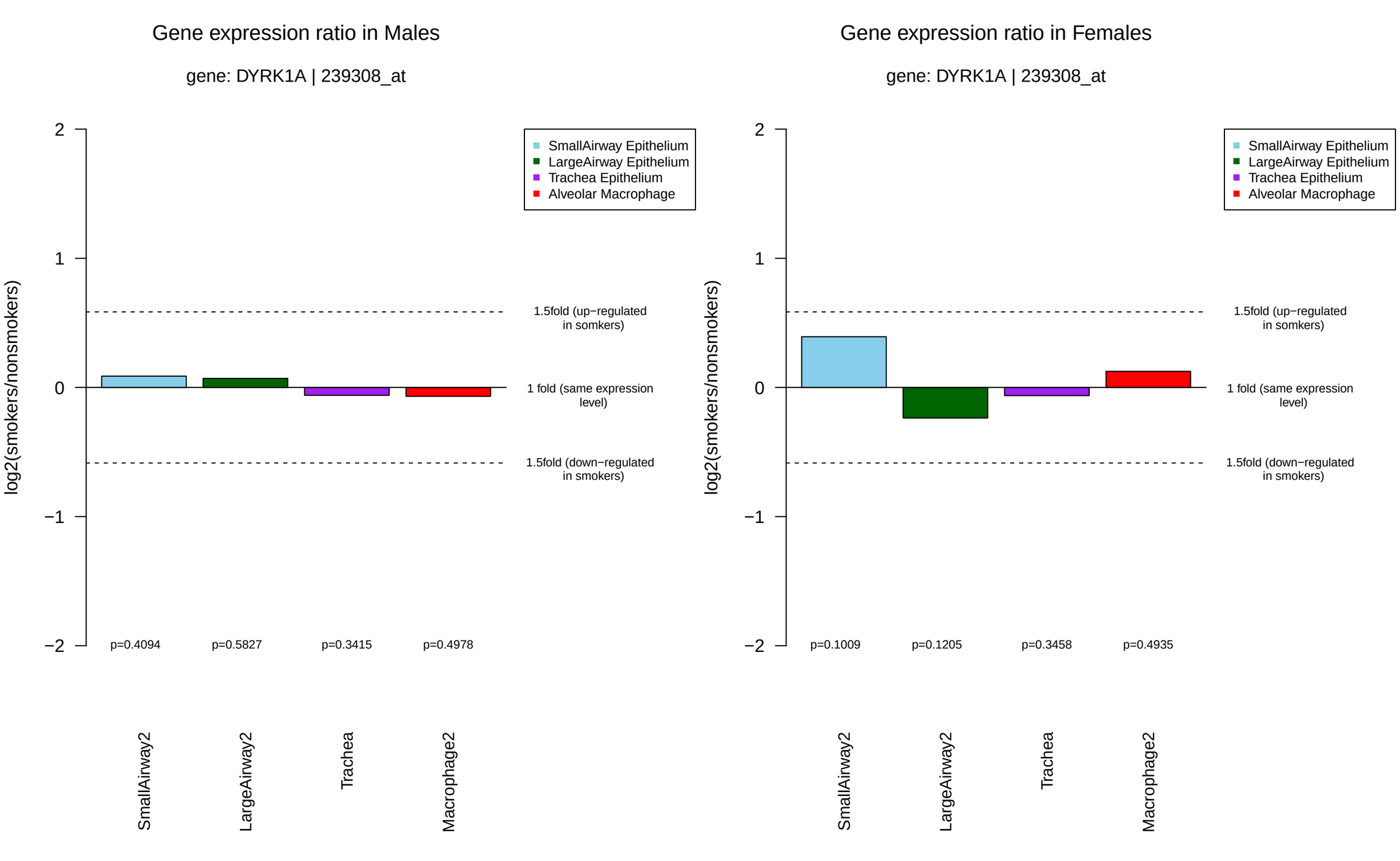
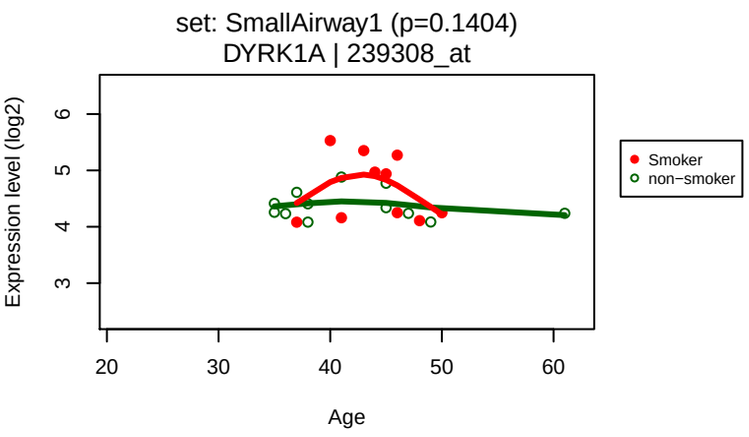
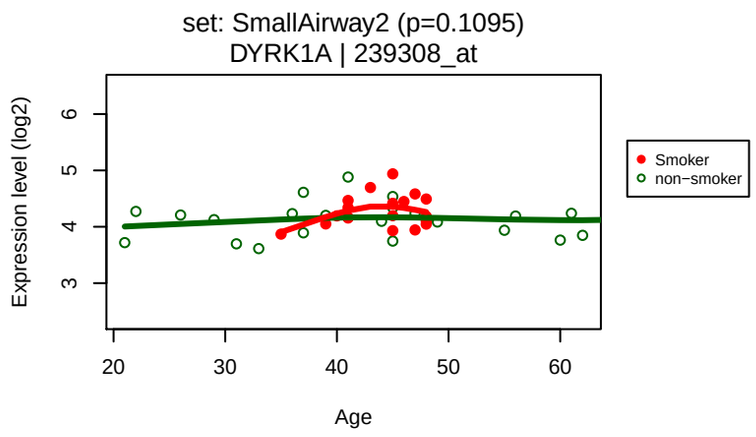
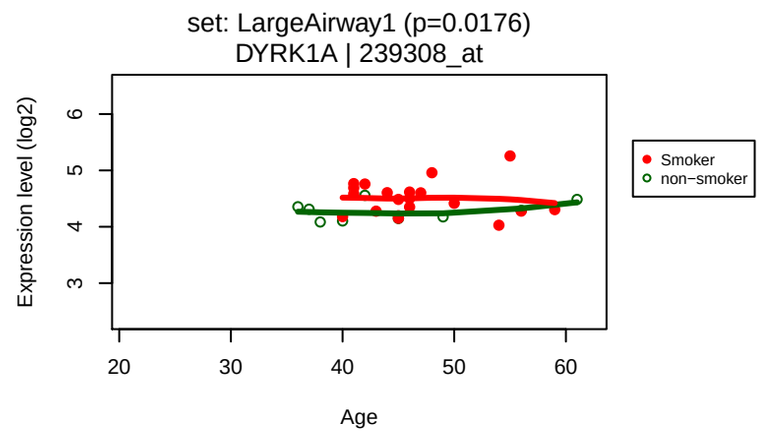

Gene expression of smokers and nonsmokers
gene: DYRK1A | 239308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1404
p=0.1095
p=0.1502
p=0.0176
p=0.8597
p=0.1597
p=0.2162
p=0.8795
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

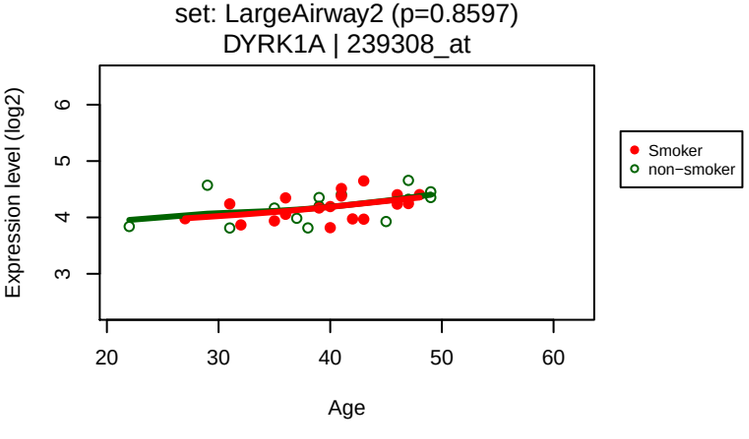

Gene expression ratio (smokers/nonsmokers)
gene: DYRK1A | 239308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1404
p=0.1095
p=0.1502
p=0.0176
p=0.8597
p=0.1597
p=0.2162
p=0.8795
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

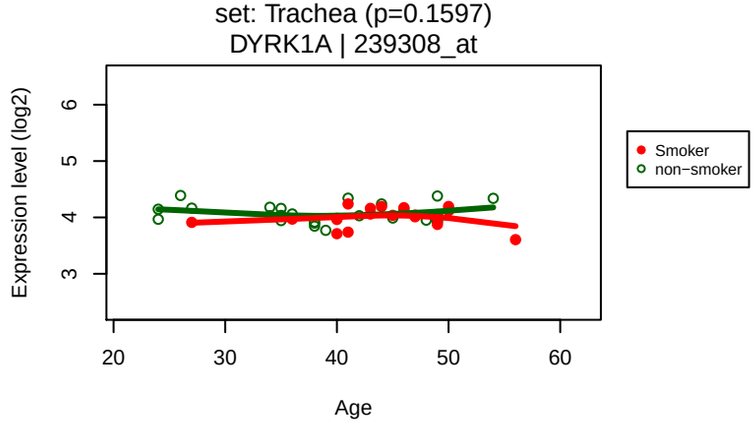

Gene expression ratio in Males
gene: DYRK1A | 239308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4094
p=0.5827
p=0.3415
p=0.4978
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DYRK1A | 239308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1009
p=0.1205
p=0.3458
p=0.4935
SmallAirway2
LargeAirway2
Trachea
Macrophage2

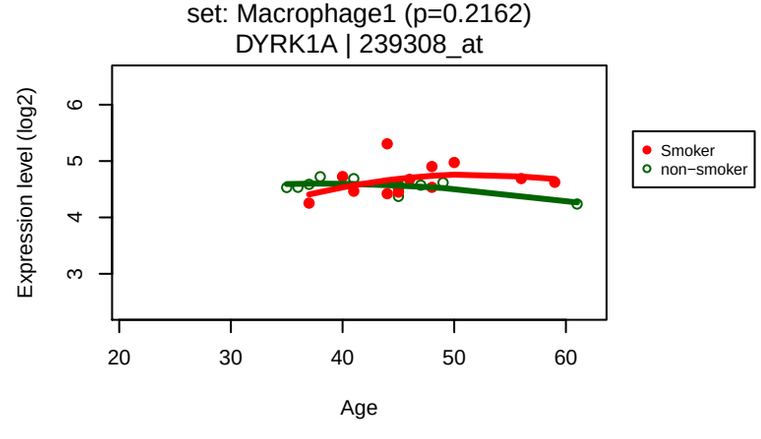

set: SmallAirway1 (p=0.1404)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker

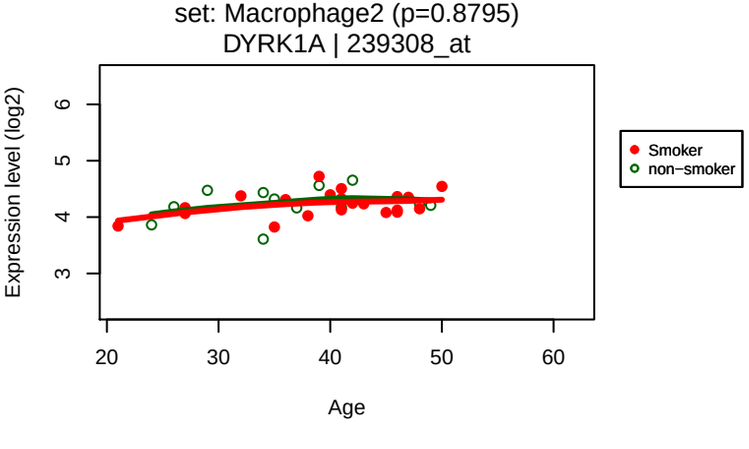

set: SmallAirway2 (p=0.1095)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker

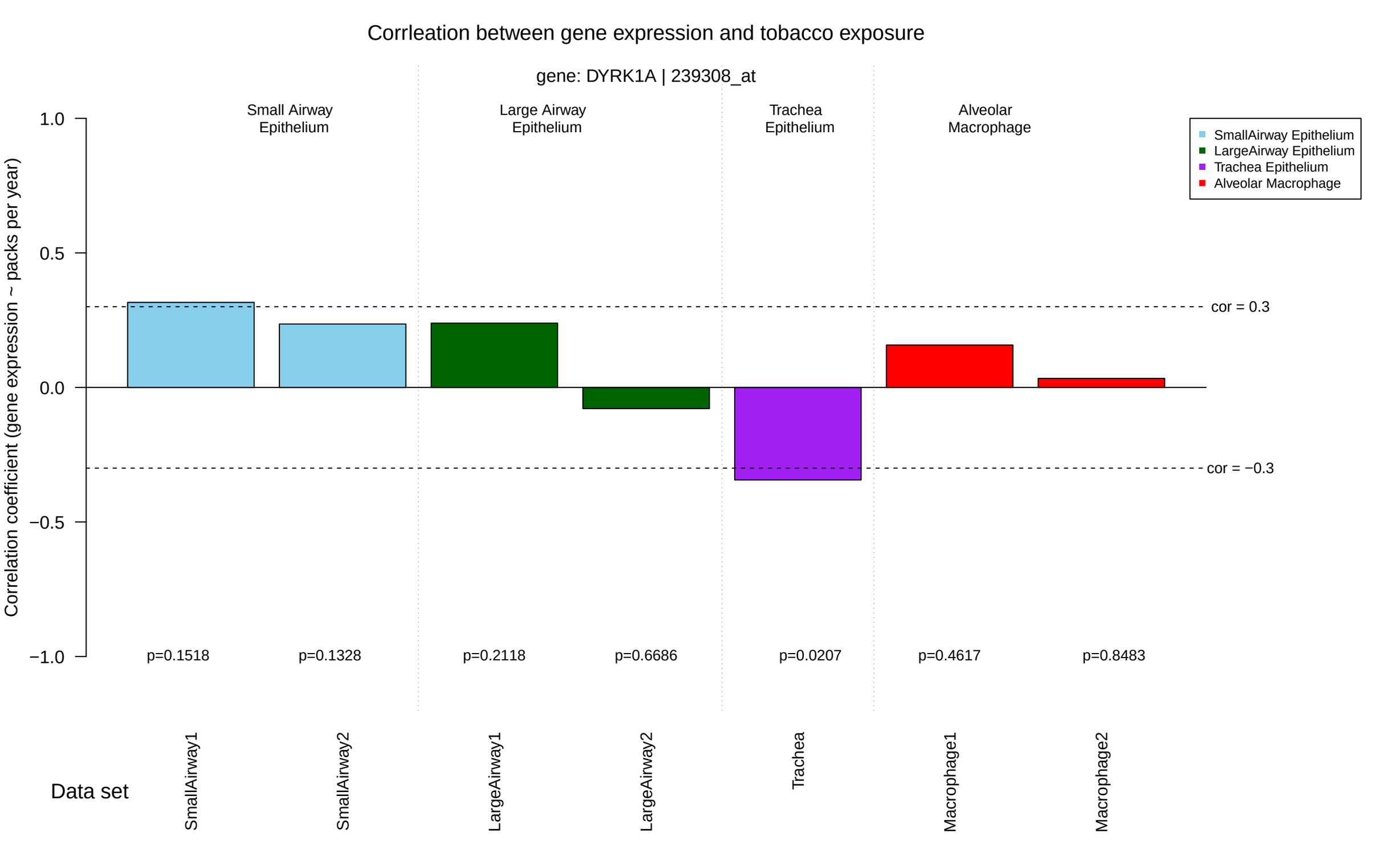

set: LargeAirway1 (p=0.0176)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8597)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1597)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2162)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8795)
DYRK1A | 239308_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DYRK1A | 239308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1518
p=0.1328
p=0.2118
p=0.6686
p=0.0207
p=0.4617
p=0.8483
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2